

ADSM — Adaptive Decision Stability Module

Parent Standard: Operational Decision Integrity Standard (ODIS)

Category: Governance & Enforcement

Subcategory: Adaptive Decision Stability

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Compatibility: OOF Methodology OS · Operational Decision Integrity Standard (ODIS) · Runtime Integrity Standard (RIS) · Operational Context Integrity Standard (OCIS) · Operational Intent Integrity Standard (OIIS) · Operational Evidence & Auditability Standard (OEAS) · INTEGROS® — Integrity Standard · Autonomous Runtime Systems · Adaptive AI Infrastructures

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition

Adaptive Decision Stability Module (ADSM) defines the structural conditions under which adaptive runtime decisions, dynamic decision evolution, distributed adaptive coordination, and consequence-bearing adaptive decision states remain materially stable, traceable, and operationally governable across adaptive runtime environments.

A system satisfies ADSM only if:

- adaptive runtime decisions remain materially stable
- dynamic decision evolution preserves operational alignment
- distributed adaptive coordination remains coherent
- consequence-bearing adaptive decisions remain operationally governable
- adaptive decision drift does not destabilize operational continuity

A system that preserves runtime execution while adaptive operational decisions materially destabilize does not satisfy ADSM.

Module Operational Role

ADSM defines the adaptive decision stability layer of ODIS by preserving governance-valid adaptive runtime decisions across operational environments.

Module Operational Space

ADSM governs the adaptive decision stability space of ODIS by preserving dynamic decision continuity, distributed adaptive coordination, operational alignment stability, consequence-bearing adaptive governance, and runtime decision coherence across operational systems.

Module Function

The module applies wherever systems must preserve:

- adaptive runtime decision stability
- dynamic decision continuity
- distributed adaptive coordination
- consequence-bearing adaptive governance
- operational alignment continuity
- governance-valid decision evolution

Its function is to ensure that adaptive operational decisions remain materially stable strongly enough to preserve runtime decision legitimacy across execution environments.

Minimum Implementation Framework

1. Define the Adaptive Decision Object

The organization must define which adaptive operational decision structures require stability governance.

This may include:

- adaptive runtime decision systems
 - dynamic orchestration routing
 - distributed autonomous coordination
 - realtime operational adaptation
 - escalation-sensitive adaptive actions
 - consequence-bearing runtime decisions
 - operational-critical adaptive pathways
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2. Define Adaptive Decision Stability Conditions

The system must define the conditions under which adaptive operational decisions remain materially stable and operationally aligned.

This includes:

- dynamic decision continuity
 - adaptive coordination coherence
 - operational alignment stability
 - runtime decision traceability
 - governance-valid adaptive evolution
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3. Define Adaptive Decision Drift Detection Logic

The system must define how materially unstable adaptive operational decisions or adaptive decision drift is identified.

This may include:

- adaptive operational divergence
 - runtime decision instability
 - distributed coordination fragmentation
 - escalation-sensitive adaptive corruption
 - consequence-bearing decision misalignment
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4. Define Operational Response or Governance Logic

The system must define governance logic for materially unstable adaptive-decision conditions.

Governance response may include:

- adaptive decision stabilization
 - coordination synchronization
 - escalation review activation
 - runtime adaptation restriction
 - operational reconstruction
 - operational invalidation where required
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5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of adaptive-decision continuity and adaptive-decision drift states. A system must not remain decision-valid if adaptive runtime decisions materially destabilize while systems continue assuming governance-valid operational alignment remains preserved.

Use Case 1 — Adaptive AI Orchestration

Environment

Scenario

An adaptive AI infrastructure continuously evolves runtime operational decisions across distributed orchestration systems and autonomous execution environments.

Application

ADSM preserves governance-valid adaptive decision continuity through runtime stabilization governance and distributed coordination alignment.

Result

The organization gains stronger adaptive runtime predictability and reduced hidden adaptive-decision instability across distributed AI systems.

Use Case 2 — Autonomous Industrial Runtime

Infrastructure

Scenario

An autonomous industrial infrastructure continuously adapts realtime operational decisions across distributed execution systems and dynamic operational environments.

Application

ADSM preserves governance-valid adaptive runtime alignment through distributed coordination governance and operational decision stabilization.

Result

The environment gains stronger runtime decision continuity and reduced adaptive operational divergence across autonomous operational systems.

Canonical Closing Statement

If adaptive runtime decisions cannot remain materially stable across operational environments, systems may preserve execution continuity while operational decision legitimacy progressively diverges from governance- valid alignment.

Related Documents

→ [Operational Decision Integrity Standard - \(ODIS\)](#).

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